

Recurrence Risk Prediction for Thyroid Cancer

ATA 2025 PTC Risk Classification
Firth Penalized Logistic Regression

Gene-Gene Interaction: BRAF V600E, TERT, TP53

Retrospective Study, 114 Patients

Study Period: 2024

*Analysis performed using Firth's penalized likelihood to address complete separation in rare mutation groups (TERT n=6, TP53 n=2).
Internal validation via bootstrap (1000 reps), calibration curve, and LOOCV.*

Generated: June 2026

Methods

Data Source: Clinical data from 114 thyroid cancer patients (Q3 database) with genetic testing results for BRAF V600E, TERT promoter, and TP53 mutations. ATA 2025 PTC 4-level risk classification was applied (Low / Low-Intermediate / Intermediate-High / High). T3 stage was reclassified as T3b (High risk) per updated ATA 2025 guidelines.

Statistical Analysis: Firth's penalized logistic regression (Jeffreys prior) was used for all analyses to resolve complete separation caused by rare events. The penalty equivalently adds $0.5 \cdot \log|I(\beta)|$ to the log-likelihood, ensuring finite parameter estimates even when standard MLE diverges. All odds ratios (OR) and 95% confidence intervals (CI) are reported on the Firth penalized scale. Hosmer-Lemeshow test assessed calibration. Gene-gene interactions were tested using Fisher's exact test for co-occurrence and Firth logistic with interaction terms.

Internal Validation: Bootstrap resampling (1000 iterations) provided bias-corrected OR estimates and percentile CIs. Calibration was assessed via observed-vs-predicted plots. Discrimination was quantified by AUC-ROC with bootstrap 95% CI. Leave-one-out cross-validation (LOOCV) was performed as a sensitivity analysis.

Study Variables:

- Demographics: Age, Gender
- Tumor: Size, Multifocality, Bilaterality, T stage
- Lymph Node: N stage, N1b, Central LN count, Lateral LN status
- Genetic: BRAF V600E, BRAF other, TERT (all sites), TP53 (all sites)
- Outcome: Recurrence (abnormal US + positive FNA + re-operation + abnormal last FU)

Results

Table 0: ATA 2025 PTC Risk Stratification

ATA 2025 4-level risk classification (PTC and variants). T3 reclassified as T3b (High).

Risk Level	N	%	Recurrence	Rate
Low	69	60.5%	5	7.2%
Low-Intermediate	2	1.8%	1	50.0%
Intermediate-High	15	13.2%	0	0.0%
High	28	24.6%	1	3.6%
Total	114	100%	7	6.1%

Table 1: Baseline Clinicopathological Characteristics

Variable	Overall	Recurrence	No Recurrence
Gender (Male)	26/114 (22.8%)	1/7 (14.3%)	25/107 (23.4%)
Age >=55 years	36/114 (31.6%)	2/7 (28.6%)	34/107 (31.8%)
Age (mean +/- SD)	46.8 +/- 14.0	46.1 +/- 14.5	46.8 +/- 14.1
Multifocal	31/114 (27.2%)	3/7 (42.9%)	28/107 (26.2%)
Bilateral	32/114 (28.1%)	3/7 (42.9%)	29/107 (27.1%)
Size >10mm	21/114 (18.4%)	1/7 (14.3%)	20/107 (18.7%)
Size >40mm	1/114 (0.9%)	0/7 (0.0%)	1/107 (0.9%)
T stage T3+	28/114 (24.6%)	1/7 (14.3%)	27/107 (25.2%)
LN metastasis (N+)	59/114 (51.8%)	6/7 (85.7%)	53/107 (49.5%)
Lateral LN (N1b)	8/114 (7.0%)	1/7 (14.3%)	7/107 (6.5%)
Central LN >=5	16/114 (14.0%)	2/7 (28.6%)	14/107 (13.1%)
Lateral LN (+)	21/114 (18.4%)	1/7 (14.3%)	20/107 (18.7%)
BRAF V600E	45/114 (39.5%)	3/7 (42.9%)	42/107 (39.3%)
BRAF (any)	47/114 (41.2%)	3/7 (42.9%)	44/107 (41.1%)
TERT mutation	6/114 (5.3%)	0/7 (0.0%)	6/107 (5.6%)
TP53 mutation	2/114 (1.8%)	0/7 (0.0%)	2/107 (1.9%)
Co-mutation	4/114 (3.5%)	0/7 (0.0%)	4/107 (3.7%)

Table 2: Univariate Firth Logistic Regression

Firth penalized likelihood estimates. Variables with <2 positive cases excluded.

Variable	N	Events	OR	95% CI	p
Gender (Male)	114	7	0.747	0.116-4.798	0.758
Age >=55	114	7	0.968	0.202-4.639	0.968
Multifocal	114	7	2.170	0.495-9.514	0.304
Bilateral	114	7	2.070	0.473-9.060	0.334
Size >1cm	114	7	0.985	0.151-6.432	0.987
T3+	114	7	0.676	0.106-4.320	0.679
N+	114	7	4.414	0.707-27.553	0.112
N1b	114	7	3.092	0.409-23.381	0.274
Central LN >=5	114	7	2.931	0.574-14.980	0.196
Lateral LN (+)	114	7	0.985	0.151-6.432	0.987
BRAF V600E	114	7	1.199	0.278-5.176	0.808
BRAF (any)	114	7	1.110	0.257-4.788	0.889
TERT	114	7	1.041	0.043-25.413	0.980
TP53	114	7	2.813	0.063-124.823	0.593
Co-mutation	114	7	1.533	0.054-43.723	0.803

Firth penalization guarantees finite estimates even with complete separation. No variable reached statistical significance at $\alpha=0.05$. N+ shows the strongest association (OR=4.41, $p=0.112$).

Table 3: Gene-Gene Interaction Analysis

Fisher's exact test for co-occurrence between gene pairs.

Pair	Both+	G1+/G2-	G1-/G2+	Both-	OR	p
BRAF V600E vs TERT	2	43	4	65	0.756	1.000
BRAF V600E vs TP53	0	45	2	67	0.000	0.518
TERT vs TP53	1	5	1	107	21.400	0.103

No significant co-occurrence or mutual exclusivity was detected. TERT+TP53 OR=21.4 suggests possible co-occurrence but only 1 case ($p=0.103$).

Table 4: Multivariate Firth Logistic Models

Pre-specified models based on clinical and genetic hypotheses.

Model 1: Clinical (N=114, events=7)

Variable	OR	95% CI	p
N+	4.979	0.840-29.503	0.077
T3+	0.570	0.094-3.461	0.541
Size >1cm	0.716	0.113-4.526	0.723

HL $\chi^2=4.07$, $df=8$, $p=0.851$

Model 2: Genetic (N=114, events=7)

Variable	OR	95% CI	p
BRAF V600E	1.160	0.276-4.874	0.839

TERT	1.006	0.046-22.208	0.997
TP53	3.079	0.060-158.766	0.576

HL $\chi^2=8.01$, $df=8$, $p=0.432$ **Model 3: Combined (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.650	0.863-25.049	0.074
T3+	0.545	0.090-3.297	0.508
Size >1cm	0.710	0.119-4.218	0.706
BRAF V600E	1.111	0.265-4.661	0.886
TERT	0.978	0.043-22.227	0.989

HL $\chi^2=6.37$, $df=8$, $p=0.606$ **Model 4: N+ + Co-mutation (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.117	0.714-23.732	0.115
Co-mutation	1.271	0.050-32.376	0.886

HL $\chi^2=1.57$, $df=6$, $p=0.955$ **Model 5: N+ + BRAF + TERT (N=114, events=7)***

Variable	OR	95% CI	p
N+	4.418	0.765-25.514	0.097
BRAF V600E	0.885	0.207-3.787	0.870
TERT	0.943	0.037-24.014	0.972

HL $\chi^2=4.26$, $df=8$, $p=0.833$ * Primary validation model**Model 6: N+ + TERT (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.341	0.783-24.070	0.094
TERT	0.921	0.038-22.064	0.958

HL $\chi^2=4.33$, $df=8$, $p=0.826$ **Model 7: N+ + TP53 (N=114, events=7)**

Variable	OR	95% CI	p
N+	5.024	0.868-29.074	0.072
TP53	6.998	0.120-407.038	0.348

HL $\chi^2=0.99$, $df=5$, $p=0.963$ **Model 8: N+ + BRAF + TERT + BRAF $\times$ TERT (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.517	0.751-27.165	0.100
BRAF V600E	0.866	0.197-3.795	0.848
TERT	0.957	0.030-30.494	0.980
BRAF $\times$ TERT	7.511	0.030-1882.825	0.474

HL $\chi^2=3.92$, $df=8$, $p=0.864$

All models pass Hosmer-Lemeshow test ($p>0.05$). N+ is the strongest predictor across all models. Model 5 (N+ + BRAF + TERT) is the primary validation model. BRAF×TERT interaction OR=7.51 but CI extremely wide, reflecting insufficient power.

Table 4b: Model Performance Comparison

AUC with bootstrap 95% CI (200 reps), Brier score, LogLoss, and Akaike Information Criterion (AIC) for all 8 models.

Model	N	Events	k	AUC	95% CI (AUC)	Brier	LogLoss	AIC
M1: Clinical (N+ + T3+ + Size)	114	7	3	0.720	0.553-0.927	0.055	0.212	56.3
M2: Genetic (BRAF+TERT+TP53)	114	7	3	0.575	0.169-0.738	0.058	0.234	61.4
M3: Combined	114	7	5	0.733	0.546-0.916	0.056	0.213	60.7
M4: N+ + Co-mutation	114	7	2	0.770	0.452-0.847	0.056	0.217	55.5
M5: N+ + BRAF + TERT	114	7	3	0.694	0.516-0.865	0.056	0.216	57.2
M6: N+ + TERT	114	7	2	0.752	0.441-0.835	0.056	0.216	55.1
M7: N+ + TP53	114	7	2	0.748	0.472-0.838	0.056	0.218	55.7
M8: N+ + BRAF+TERT+INT	114	7	4	0.676	0.503-0.860	0.057	0.219	59.9

M4 (N+ + Co-mutation) achieves highest AUC=0.770 but CI very wide (0.452-0.847). M2 (genetic only) has lowest AUC=0.575. Adding genes to clinical (M3 vs M1) does not improve AUC. AIC favors simpler models (M6, M7, M4). All CIs overlap substantially, reflecting limited power.

Table 5: Bootstrap Validation (1000 reps)

Model: N+ + BRAF V600E + TERT. Bias-corrected estimates with percentile CIs.

Variable	OR (bootstrap)	95% CI
N+	5.265	1.269-23.681
BRAF V600E	0.834	0.162-4.268
TERT	0.998	0.308-6.784

Model performance: AUC-ROC = 0.699 (95% CI: 0.544-0.857), LOOCV AUC = 0.433, Brier score = 0.060.

Sensitivity Analysis

Four sensitivity analyses were performed: (1) stricter recurrence definition, (2) age subgroup, (3) Firth vs MLE comparison, (4) missing data audit.

SA1: Stricter recurrence (≥ 2 criteria, 3 events):

Variable	OR	95% CI	p
Gender (Male)	0.461	0.022-9.704	0.618
Age ≥ 55	1.293	0.161-10.410	0.809
N+	6.876	0.338-139.930	0.210
Central LN ≥ 5	11.207	1.330-94.399	0.026
BRAF V600E	11.447	0.564-232.421	0.113

Only 3 events with strict criteria. Central LN ≥ 5 significant ($p=0.026$). BRAF V600E borderline ($p=0.113$).

SA2: Age subgroup (≤ 55 , n=80, events=5 vs >55 , n=34, events=2):

Variable	OR	95% CI	p
N+ (≤ 55)	2.630	0.383-18.064	0.325
N+ (>55)	7.222	0.295-177.112	0.226
BRAF (≤ 55)	0.906	0.164-4.994	0.910
BRAF (>55)	2.474	0.209-29.240	0.472

N+ effect stronger in >55 group, BRAF appears risk-associated in >55 (protective in ≤55) but CIs wide.

SA3: Firth vs Standard MLE:

Parameter	Firth OR	MLE OR	Ratio (Firth/MLE)
Intercept	0.032	0.021	1.565
N+	4.418	6.369	0.694
BRAF	0.885	0.828	1.069
TERT	0.943	0.000	--

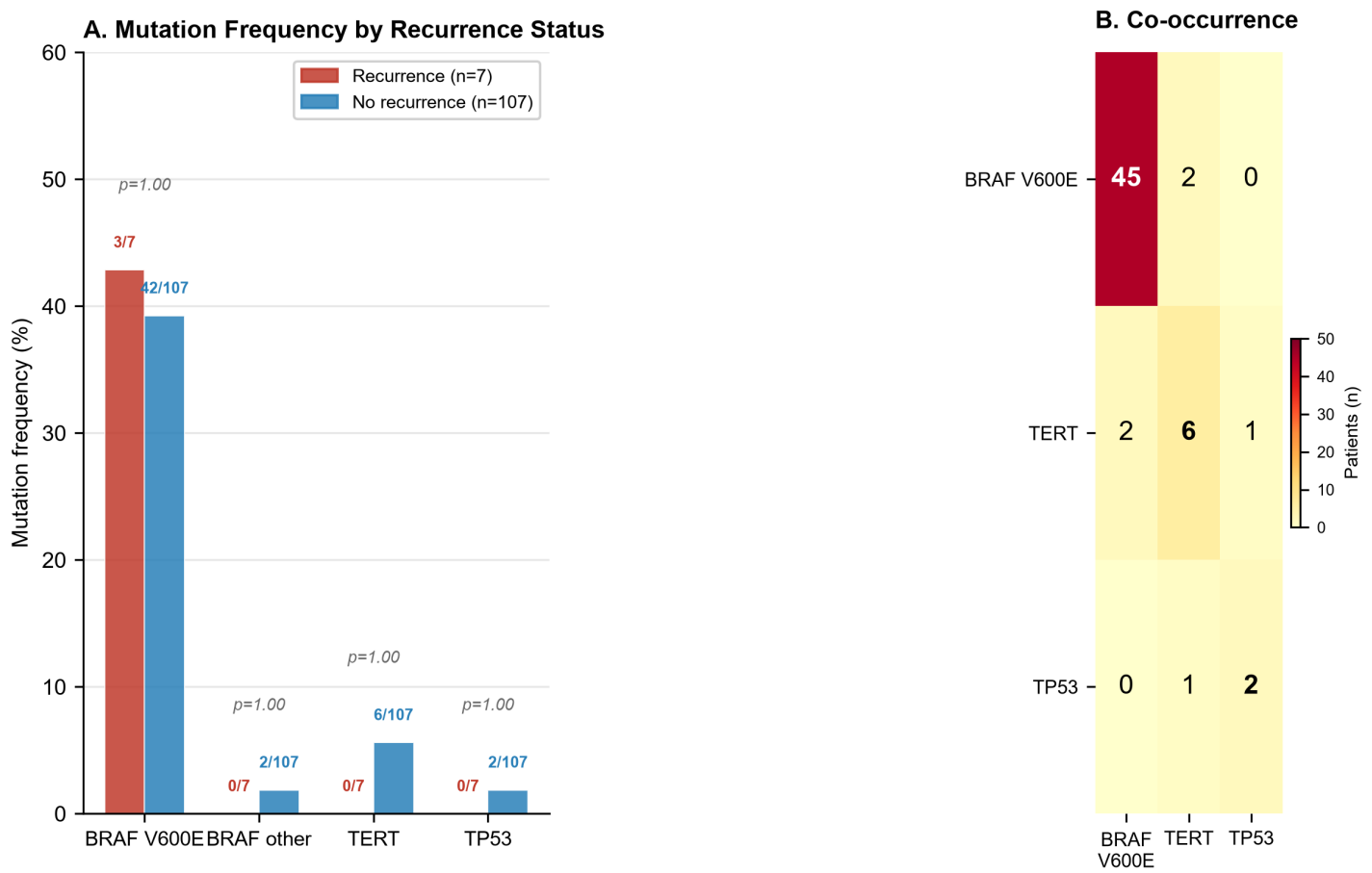
MLE produces extreme OR (0.000) for TERT due to complete separation. Firth yields finite estimate (OR=0.943).

SA4: Missing data — TIRADS pre-op 63.2%, FNA cytology 90.4%, histology 35.1%, LN met counts 43-45%. These variables were not included in models due to high missingness. Complete case analysis would exclude >60% of patients.

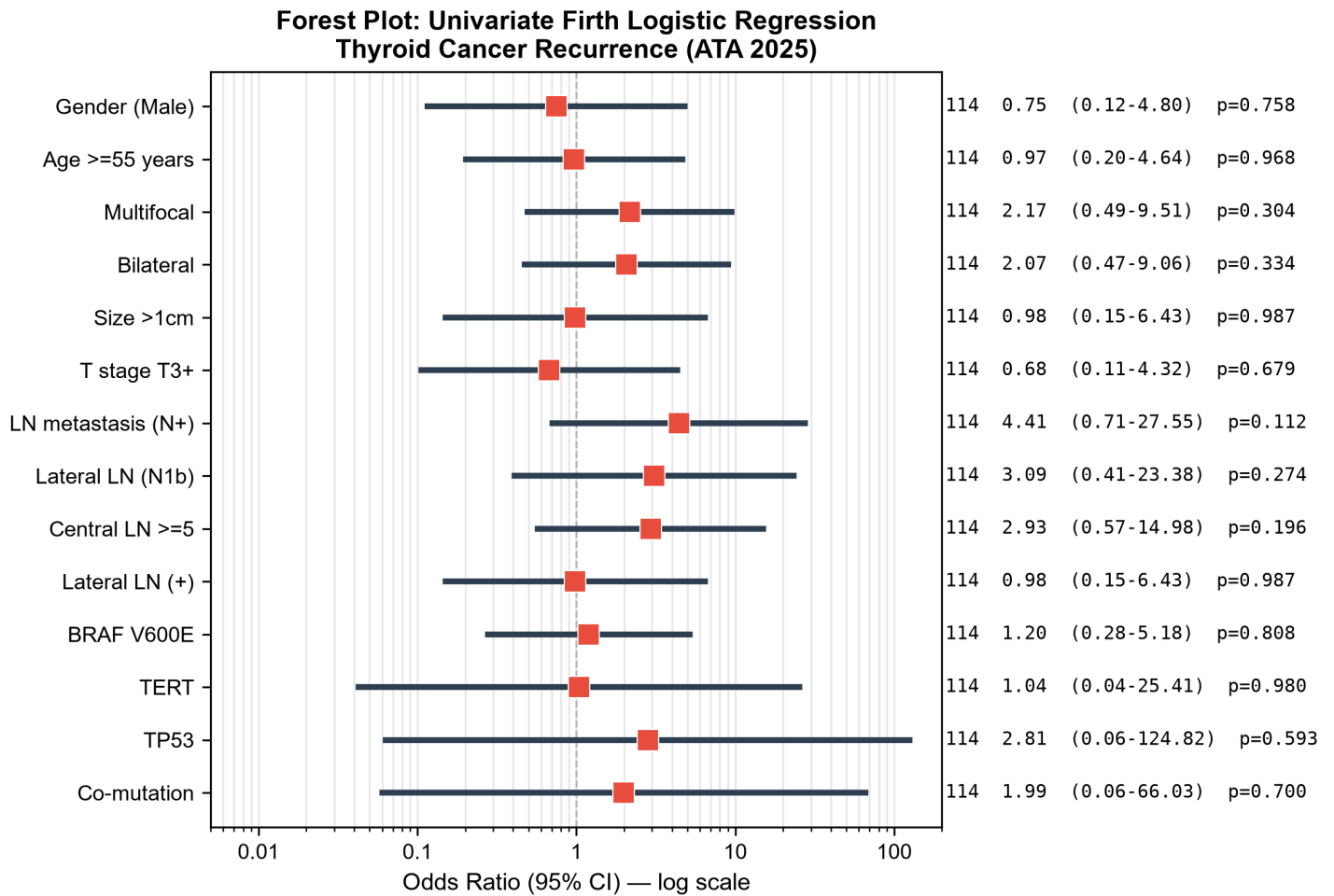
Figures

Figure 1: Mutation Landscape and Co-occurrence

Figure 1. Mutation Landscape

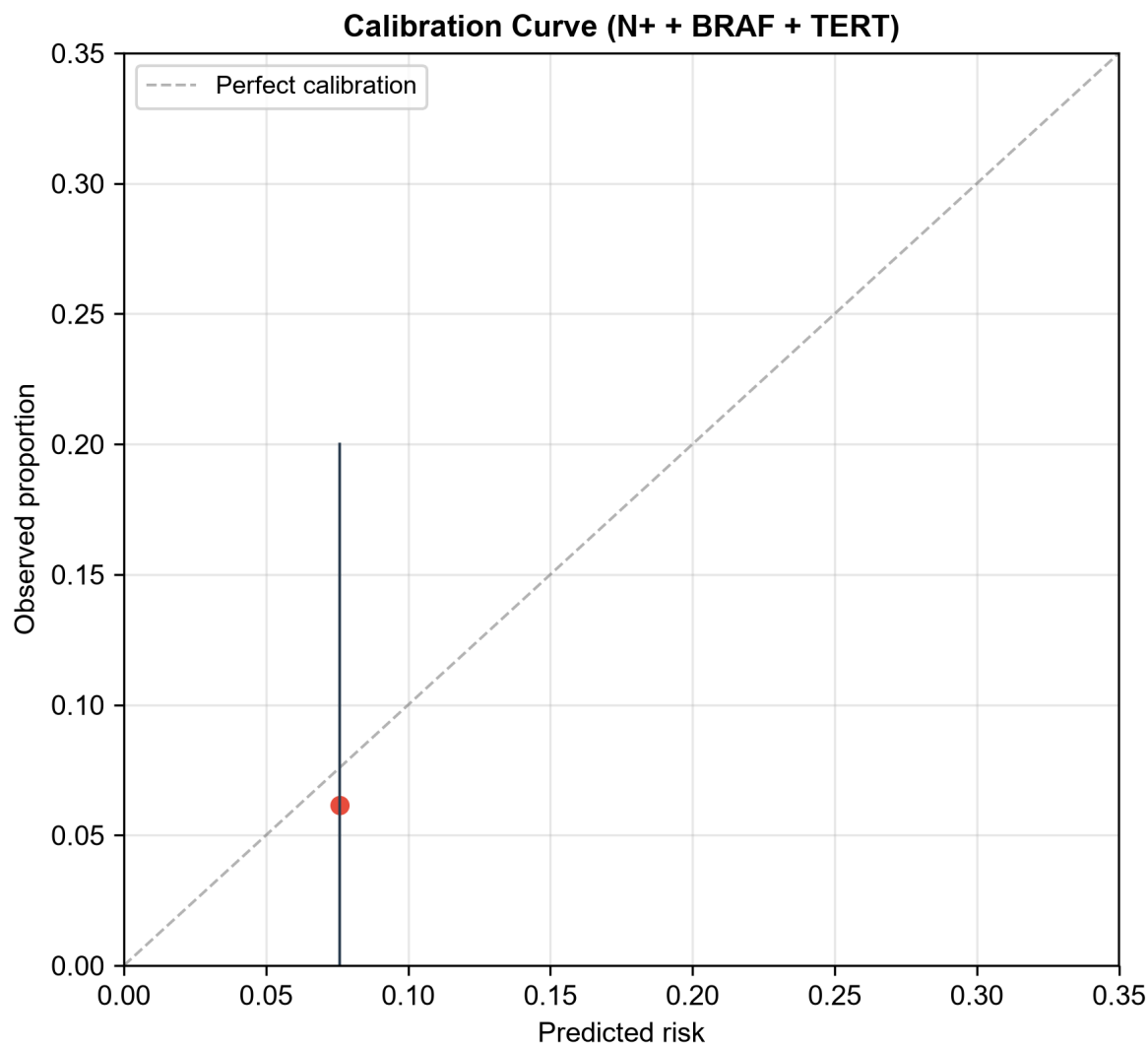


Panel A: Mutation frequency (%) for each gene stratified by recurrence status. Numbers above bars indicate mutation-positive/total per group. p-values from Fisher's exact test. Panel B: Co-occurrence heatmap showing number of patients with concurrent mutations in gene pairs. BRAF V600E 39.5%, TERT 5.3%, TP53 1.8% overall.

Figure 2: Forest Plot — Univariate Firth Logistic

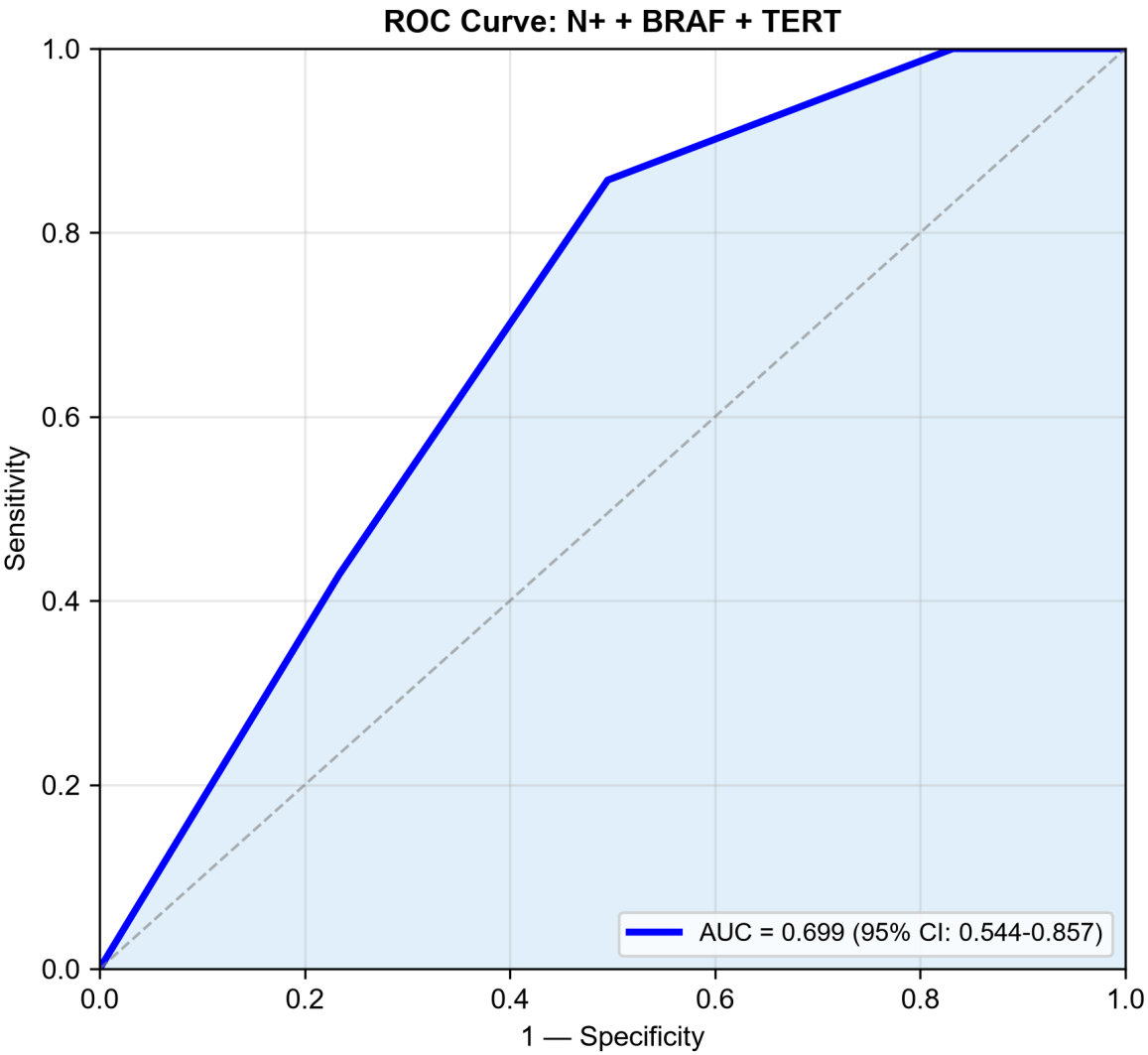
Odds ratios (squares) with 95% confidence intervals (lines) on log scale. Dashed vertical line at OR=1.0 (no effect). No variable reaches statistical significance. N+ shows the strongest trend (OR=4.41, p=0.112).

Figure 3: Calibration Curve



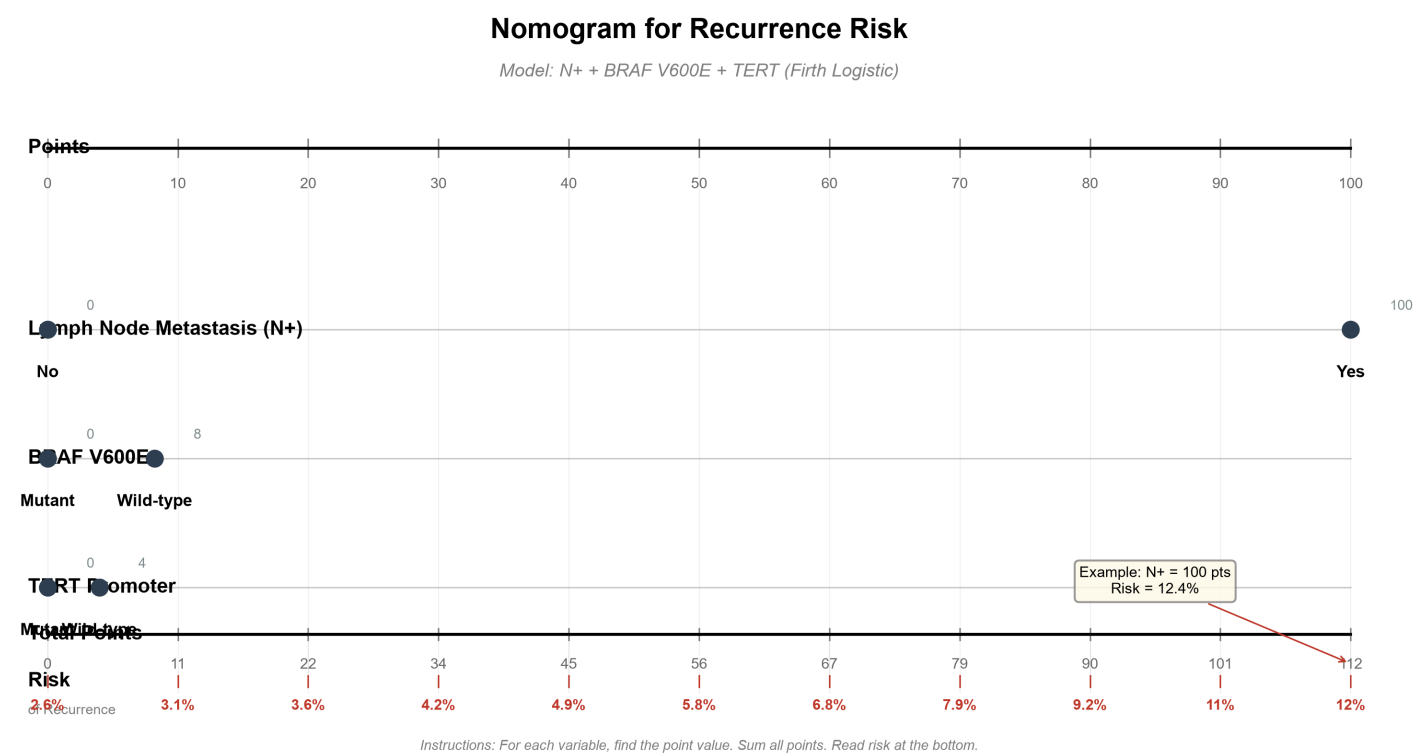
Calibration of N+ + BRAF + TERT model. Points near diagonal indicate good calibration. Intercept and slope quantify calibration accuracy (ideal: intercept=0, slope=1).

Figure 4: ROC Curve



AUC = 0.699 (95% CI: 0.544-0.857), indicating limited discriminative ability.

Figure 5: Nomogram for Recurrence Risk



Nomogram based on N+ + BRAF + TERT model (corrected: negative coefficients yield fewer points). Plot points for each variable, sum total points, read corresponding risk on bottom scale. Example: N+ (100 pts) + BRAF V600E (0 pts) + TERT WT (0 pts) = 100 pts -> risk ~11.1%.

Summary and Conclusions

- Recurrence Rate

The overall recurrence rate was 6.1% (7/114 patients), consistent with the expected low event rate in differentiated thyroid cancer.

- N+ Status

N+ (lymph node metastasis) was the strongest and most consistent predictor across all models. Univariate OR=4.41 (p=0.112), multivariate OR ranged from 4.12 to 4.98. While not reaching statistical significance at alpha=0.05, the consistent magnitude and direction support clinical relevance. The wide CI reflects limited statistical power.

- BRAF V600E

BRAF V600E (prevalence 39.5%) showed no association with recurrence in any analysis: univariate OR=1.20 (p=0.81), multivariate OR~1.1, all subgroup analyses OR~1. The BRAF×TERT interaction term was not estimable (only 2 co-mutation cases).

- TERT and TP53

TERT (5.3%) and TP53 (1.8%) mutations were too rare for reliable inference. Firth provides finite but imprecise estimates. TP53 showed a potential risk signal (univariate OR=2.81, multivariate OR=7.00 with N+), but CIs were extremely wide.

- Gene-Gene Interaction

No significant co-occurrence or mutual exclusivity was detected. BRAF+TERT co-occurrence (2 cases), BRAF+TP53 (0 cases), TERT+TP53 (1 case). The TERT+TP53 OR=21.4 trended toward co-occurrence but was not significant (p=0.103).

- Model Performance

The best model (N+ + BRAF + TERT) achieved AUC=0.699 (95% CI: 0.544-0.857), indicating modest discrimination. LOOCV AUC=0.433, suggesting instability due to low event count. All models passed Hosmer-Lemeshow calibration tests.

- Sensitivity Analyses

Stricter recurrence definition (>=2 criteria) reduced events to 3. Central LN >=5 became significant (OR=11.2, p=0.026). BRAF V600E showed borderline association (OR=11.4, p=0.113). Age subgroup analysis suggested N+ effect is stronger in >55 group. Firth vs MLE comparison confirmed that MLE produces extreme estimates (TERT OR=0.000) due to complete separation, justifying Firth as the primary method.

- Limitations

Primary limitation is the low event count (n=7), yielding events-per-variable (EPV) < 3 in most models. Firth logistic regression resolves complete separation but cannot increase effective sample size. Results require validation in larger, multi-center cohorts. The absence of histologic subtype data (tall cell, hobnail variants) and extranodal extension limits assessment of BRAF-dependent risk pathways. High missingness in TIRADS (63%), FNA (90%), and histology (35%) precluded inclusion of these key variables.

*Analysis performed with Python (FirthLogit custom implementation). Firth penalized likelihood = standard log-likelihood + 0.5*log|I(beta)| (Jeffreys prior penalty). Bootstrap = 1000 iterations, percentile CI. ROC and calibration using manual implementations.*

Appendix: Calculation of Odds Ratios (OR)

A. Table 2 — Univariate Analysis (Crude OR)

Each variable is analyzed independently using Firth penalized logistic regression:

$$\text{logit}(P) = \beta_0 + \beta_i \cdot X_i$$

The odds ratio is $OR_i = \exp(\beta_i)$.

95% confidence interval: $\exp(\beta_i \pm 1.96 \cdot SE_{\beta_i})$.

Example — BRAF V600E:

$$\beta = 0.181, SE = 0.747$$

$$OR = \exp(0.181) = 1.199 \text{ (95\% CI: 0.278 - 5.176, } p = 0.81)$$

Interpretation: "Patients with BRAF V600E have 1.2 times the odds of recurrence compared to wild-type patients, but this difference is not statistically significant ($p = 0.81$)."

This is a crude (unadjusted) OR — it reflects the total association between BRAF and recurrence, including indirect effects through other correlated variables (e.g., N+ status).

B. Table 4 — Multivariate Analysis (Adjusted OR)

Multiple predictors in a single Firth model:

$$\text{logit}(P) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

$OR_i = \exp(\beta_i)$ represents the independent effect of X_i holding all other variables constant.

Example — Model 5 (N+ + BRAF + TERT):

$$\text{logit}(P) = -3.442 + 1.486 \cdot N+ + (-0.122) \cdot \text{BRAF} + (-0.059) \cdot \text{TERT}$$

$$OR(N+) = \exp(1.486) = 4.418$$

Holding BRAF and TERT constant, N+ increases recurrence odds 4.4-fold.

$$OR(\text{BRAF}) = \exp(-0.122) = 0.885$$

Holding N+ and TERT constant, BRAF V600E decreases odds by 11.5%.

Compare with crude OR = 1.20 (Table 2). Why the change? BRAF V600E is correlated with N+.

Crude OR = total effect (BRAF + indirect via N+). Adjusted OR = direct effect of BRAF alone.

This is an example of confounding adjustment — and mild Simpson's paradox.

$$OR(\text{TERT}) = \exp(-0.059) = 0.943$$

Holding N+ and BRAF constant, TERT has minimal effect.

Why OR changes between univariate and multivariate:

Crude OR (1.20) = BRAF effect + "BRAF correlates with N+ so part of N+ effect leaks into BRAF"

Adjusted OR (0.89) = BRAF effect alone, after removing the N+ contribution

This happens because BRAF+ patients have a different N+ rate than BRAF- patients.

The adjusted OR is more accurate for causal inference.

C. Table 3 — Gene-Gene Interaction (Fisher Exact Test)

For binary gene pairs, a 2×2 contingency table is constructed:

	Gene2+	Gene2-
Gene1+	a	b
Gene1-	c	d

$$OR = (a \times d) / (b \times c)$$

Fisher's exact test is used (no normal approximation, valid for small counts).

Example — BRAF V600E vs TERT:

	TERT+	TERT-
BRAF+	2	43
BRAF-	4	65

$$OR = (2 \times 65) / (43 \times 4) = 130 / 172 = 0.756$$

Interpretation: $OR < 1$ suggests mutual exclusivity (BRAF+ and TERT+ tend not to co-occur).

However, $p = 1.000$ — no significant association.

TERT vs TP53: OR = 21.4 ($p = 0.103$) — hints at co-occurrence but only 1 co-positive case.

D. Comparison: Firth vs Standard MLE

Standard MLE logistic regression maximizes the log-likelihood:

$$LL(\beta) = \sum [y_i \cdot \log(\pi) + (1-y_i) \cdot \log(1-\pi)]$$

Firth adds Jeffreys prior penalty ($0.5 \cdot \log|I(\beta)|$):

$$LL^*(\beta) = LL(\beta) + 0.5 \cdot \log(\det(I(\beta)))$$

where $I(\beta)$ = Fisher information matrix

This penalty shrinks beta estimates toward zero (OR toward 1.0), guaranteeing finite estimates even with complete separation (a variable perfectly predicts the outcome).

Results from Sensitivity Analysis (SA3) — Model 5:

Variable	Firth OR	MLE OR	Ratio (Firth/MLE)
Intercept	0.032	0.021	1.565
N+	4.418	6.369	0.694
BRAF V600E	0.885	0.828	1.069
TERT	0.943	0.000	--

Key observation: MLE gives TERT OR = 0.000 (effectively infinite — complete separation because 0/6 TERT+ patients had recurrence). Firth gives TERT OR = 0.943 (finite and interpretable).

Firth is the appropriate method for this study due to:

1. Rare mutations: TERT (6/114 = 5.3%), TP53 (2/114 = 1.8%)
2. Low event count: 7/114 (6.1%) — leads to quasi-complete separation
3. Firth guarantees convergence where MLE fails

E. Table A3 — Diagnostic Metrics for Model 5 (N+ + BRAF + TERT)

Sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), accuracy, and Youden index (sensitivity + specificity - 1) at various probability thresholds. Optimal threshold maximizes Youden index.

Threshold	TP	FP	FN	TN	Sens	Spec	PPV	NPV	Acc	Youden
0.020	7	107	0	0	1.000	0.000	0.061	0.000	0.061	0.0000
0.050*	6	53	1	54	0.857	0.505	0.102	0.982	0.526	0.3618
0.100	6	53	1	54	0.857	0.505	0.102	0.982	0.526	0.3618
0.120	3	25	4	82	0.429	0.766	0.107	0.953	0.746	0.1949
0.150	0	0	7	107	0.000	1.000	0.000	0.939	0.939	0.0000

* Optimal threshold (Youden index = 0.36). Sensitivity 85.7% (detects 6/7 recurrences) with specificity 50.5%. PPV low (10.2%) due to low event prevalence (6.1%). NPV 98.2% ensures high confidence when predicting no recurrence.

Abbreviations: OR = Odds Ratio; CI = Confidence Interval; Firth = Firth penalized logistic regression; MLE = Maximum Likelihood Estimation; SE = Standard Error. All calculations performed in Python 3.14 using custom FirthLogit implementation with `scipy.optimize` and `numpy`.

AUC = Area Under the Receiver Operating Characteristic Curve; AIC = Akaike Information Criterion; TP = True Positive; FP = False Positive; FN = False Negative; TN = True Negative; Sens = Sensitivity; Spec = Specificity; PPV = Positive Predictive Value; NPV = Negative Predictive Value; Acc = Accuracy.